

Week 3 – Motion Control

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Task 1 – Creating maps

Based on the analysis of example maps in the `maps` directory, the meaning of individual parameters was identified. The map file contains the goal position, environment boundaries, wall definitions, and the GNSS-denied region.

A custom map of approximately 10×10 meters was created. The map includes two internal obstacles and a large GNSS-denied region simulating an indoor environment. The map is sufficiently spacious while still providing obstacles that influence trajectory design.

Task 2 – Sample paths

Three different trajectories were defined as sequences of waypoints stored in the variable `public_vars.pat`:

- Straight line – used to verify basic functionality of the controller.
- Circular arc – used to evaluate behavior in curved motion.
- Sine wave – a more complex trajectory with continuously changing curvature.

The trajectories were designed to avoid obstacles and remain inside the valid operating region.

Task 3 – Motion control

A simple target-point based algorithm (Pure Pursuit) was implemented for path following. The robot uses its pose obtained from `read_only_vars.mocap_pose` and follows the sequence of waypoints defined by the trajectory.

The control strategy works as follows:

- a target point on the path is selected,
- the heading error between the robot orientation and the direction to the target is computed,
- wheel velocities are adjusted based on this error.

The forward velocity is adapted depending on the heading error to improve stability and smoothness of motion.

Parameter influence

During parameter tuning, the following effects were observed:

- Higher control gain resulted in faster response but caused oscillations.
- Higher forward velocity decreased tracking accuracy, especially in curved segments.
- Reducing speed near the goal improved stability and prevented oscillations.

Conclusion

The implemented algorithm was able to follow different types of trajectories, including curved paths. The quality of motion control strongly depends on parameter tuning and appropriate trajectory design with respect to obstacles.

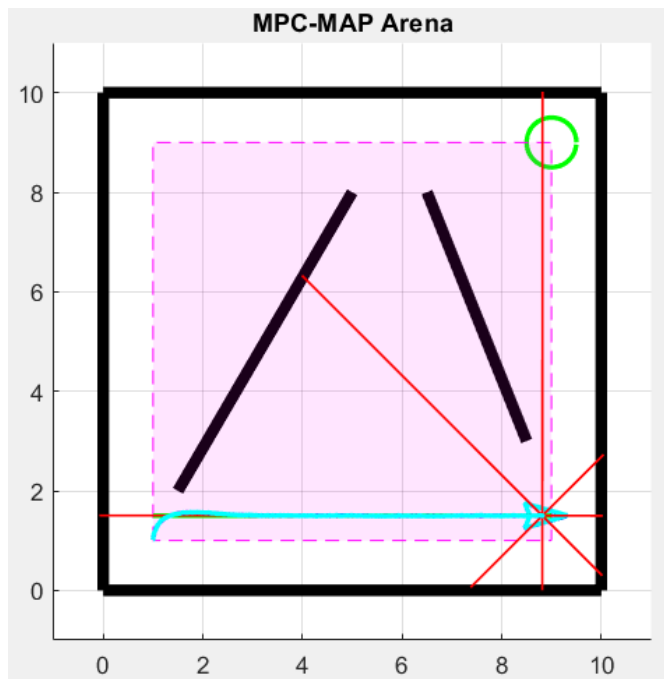


Figure 1: Straight line

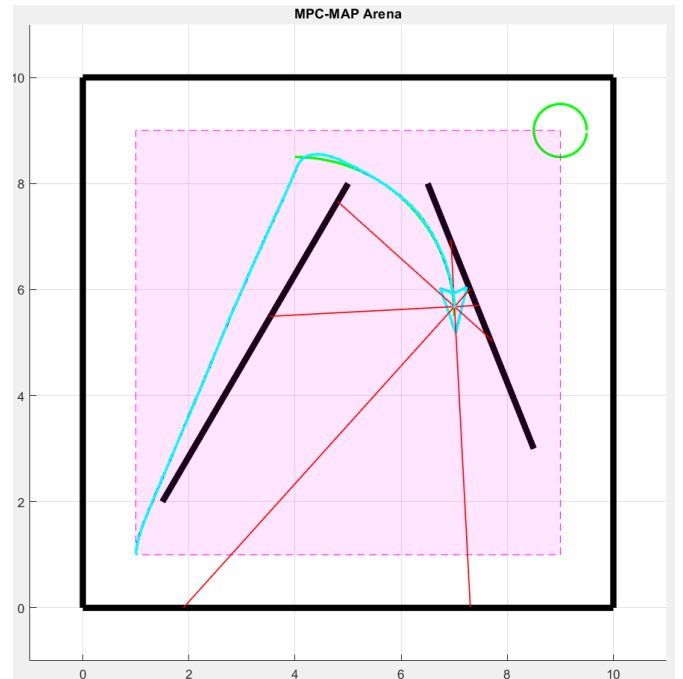


Figure 2: Circular arc

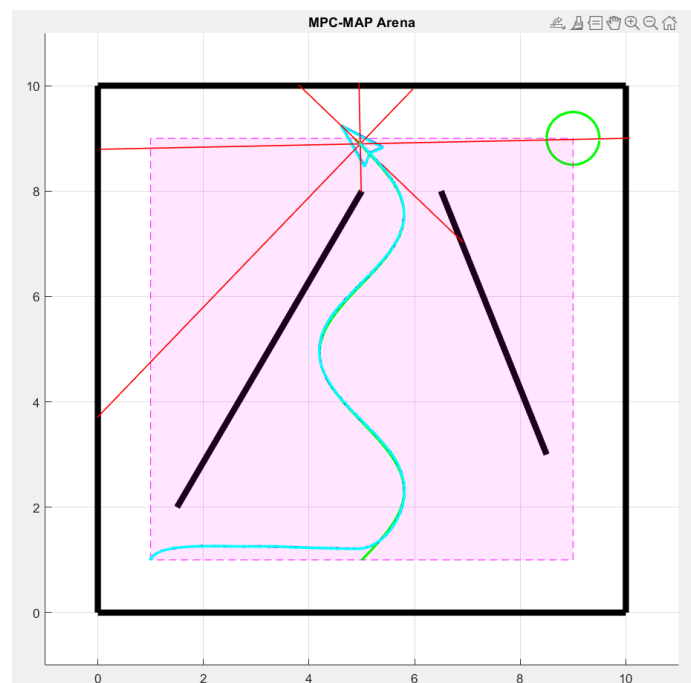


Figure 3: Sine trajectory